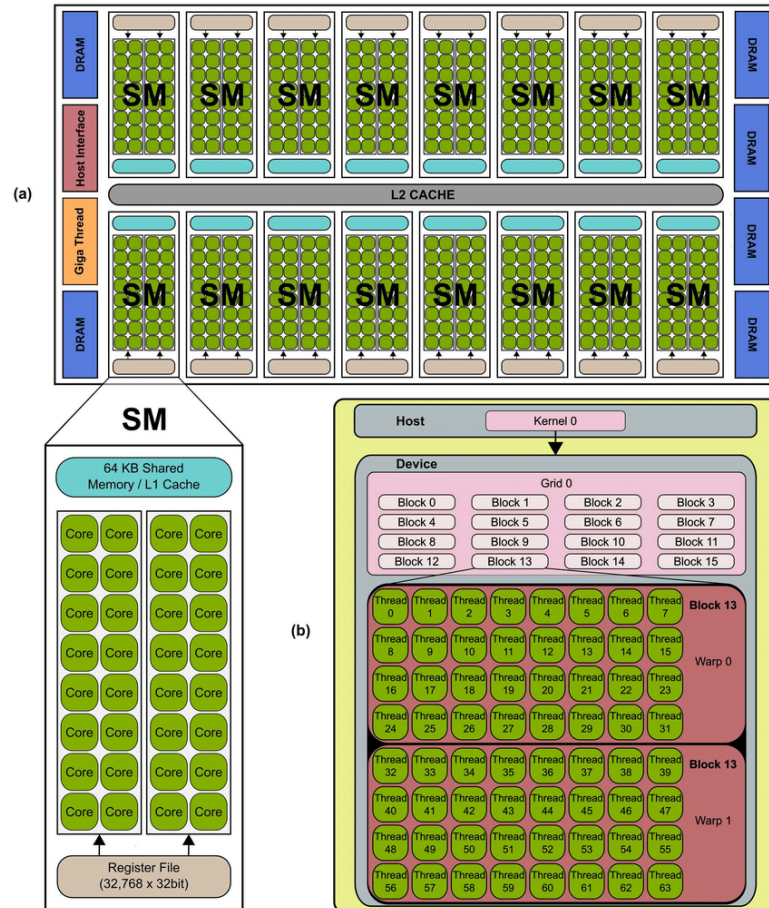


AI for AI: KernelEvolve

Feb 05 2026

GPU architecture



Kernel -> thread block -> thread

↓ ↓ ↓

Multiple SMs SM core

GPU kernels

- Basic block execution
- Implement one stage of a larger algorithm
- Examples: vecAdd, transpose, softmax
- Libraries such as cuBLAS, cuDNN are a collection of kernels
 - AMD equivalent: rocBLAS, MIOpen
- Optimized by engineers for specific hardware

Triton

- PL+compiler for custom kernels
- Mostly used for optimization or non-existent kernel
 - Kernel fusion

```
@triton.jit
def add_kernel(
    x_ptr, # *Pointer* to first input vector.
    y_ptr, # *Pointer* to second input vector.
    output_ptr, # *Pointer* to output vector.
    n_elements, # Size of the vector.
    BLOCK_SIZE: tl.constexpr, # Number of elements each program should process.
    # NOTE: `constexpr` so it can be used as a shape value.
):
    # There are multiple 'programs' processing different data. We identify which
    program
    # we are here:
    pid = tl.program_id(axis=0) # We use a 1D launch grid so axis is 0.
    # This program will process inputs that are offset from the initial data.
    # For instance, if you had a vector of length 256 and block_size of 64, the
    programs
    # would each access the elements [0:64, 64:128, 128:192, 192:256].
    # Note that offsets is a list of pointers:
    block_start = pid * BLOCK_SIZE
    offsets = block_start + tl.arange(0, BLOCK_SIZE)
    # Create a mask to guard memory operations against out-of-bounds accesses.
    mask = offsets < n_elements
    # Load x and y from DRAM, masking out any extra elements in case the input is not a
    # multiple of the block size.
    x = tl.load(x_ptr + offsets, mask=mask)
    y = tl.load(y_ptr + offsets, mask=mask)
    output = x + y
    # Write x + y back to DRAM.
    tl.store(output_ptr + offsets, output, mask=mask)
```

Benefits of kernel evolution

- Optimized kernels
- Lower human effort / faster generation
- Portability across hardware
- Lower cost?

Discussion

- What components of KernelEvolve are the most important? How would the system function without certain components?

Title: [Lec 7] Discussion